

Qatar-1b

R-0491

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_180805 · created 2026-07-20T10:01:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458335.7416 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.053
Transit depth (Rp/Rs)^2	3.14 +/- 1.87 [%]
Orbital Inclination (inc)	81.64 +/- 2.42
Airmass coefficient 1 (a1)	1.121 +/- 0.026
Airmass coefficient 2 (a2)	-0.092 +/- 0.076
Scatter in the residuals of the lightcurve fit is	2.35 %
Best Comparison Star	#1 - [556, 351]
Optimal Aperture	3.75
Optimal Annulus	13.09
Transit Duration (day)	0.048 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.053 vs 0.1463 (deviation 0.36 σ)
Duration fit vs published	0.048 d vs 0.0692 d (deviation 0.51 σ)
Mid-time O-C	3.3 min
Depth SNR	2.1
Residual scatter	2.35 %

Light curve

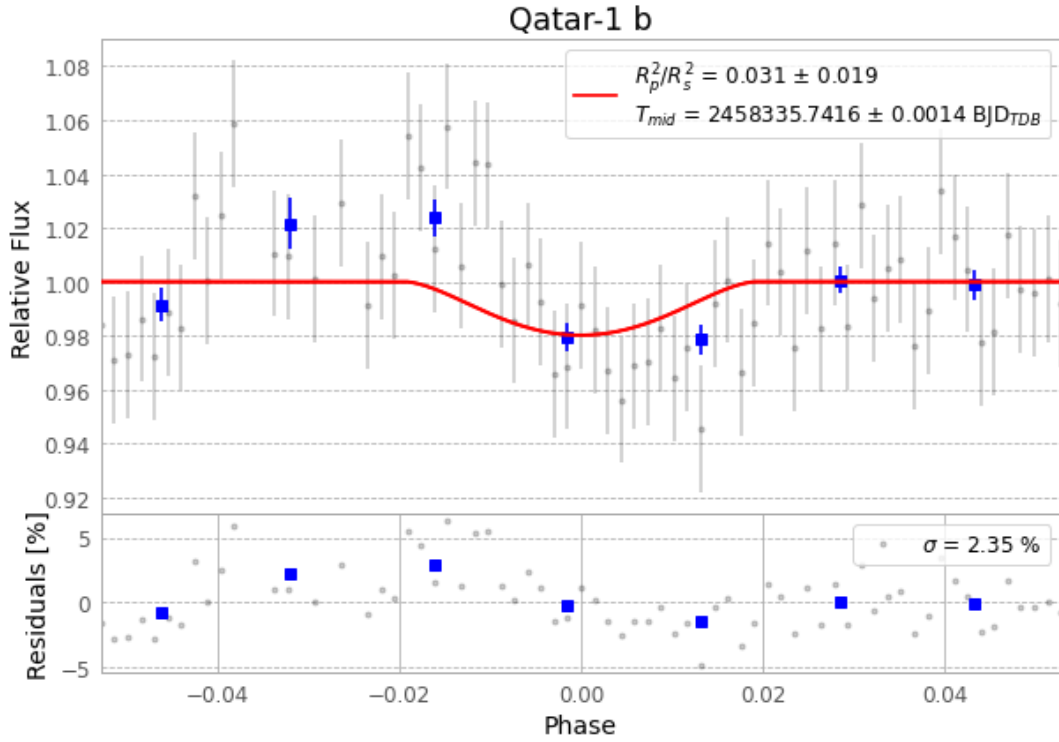


Figure 1 — FinalLightCurve_Qatar-1 b_2018-08-04.png (SHA-256 897cb5a078392c23...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 272], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 bf9c2f9ba3c046dc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

73 FITS frames (48 MB), Qatar-1180805035712.FITS → Qatar-1180805073312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-180805.log (5946a2c2c372...), FinalLightCurve_Qatar-1 b_2018-08-04.png (897cb5a07839...), FinalLightCurve_Qatar-1 b_2018-08-04.csv (d9e75c7e87e3...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 10:01Z.